

Kakoli Atri

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Education

Manipal University, Jaipur

B.Tech. CSE Specialization in IoT Engineering

Aug 2023 – May 2027

Girls' High School & College, Prayagraj

ISC Board

Apr 2009 – May 2023

Internship

AI & Robotics Intern

CIR Lab at IIT Allahabad

Prayagraj, India

June 2025 – Aug 2025

- Trained in ROS-based robotics using virtual robot simulations and controlled environments.
- Implemented robot navigation, motion control, and path planning using simulated LiDAR, IMU, and camera sensor data.
- Tested and optimized robotic behavior in dynamic simulated environments, gaining hands-on experience with robotic middleware and autonomous system workflows.

Projects

Gapwise

- Engineered a full-stack RAG platform (FastAPI, Next.js, Qdrant) using a 384-dimension embedding pipeline for AI-driven gap analysis and quiz generation across 5 dashboard modules.
- Implemented SSE streaming with Groq LLaMA 3.3 70B for real-time quiz generation and instant feedback.
- Diagnosed a 512MB memory constraint and re-architected the backend as a Docker container on HFace Spaces, scaling to 16GB via git subtree split.

Multi-Cloud CI/CD Deployment Pipeline

- Architected a DevOps pipeline for automated deployment across AWS and Azure using Jenkins and GitHub Actions.
- Built end-to-end CI/CD workflows for containerized deployments on Kubernetes with DNS-based traffic routing.
- Implemented health checks for high availability, zero downtime, and seamless fail over.

AI-Powered Collaborative Travel Planner

- Built a full-stack MERN app with real-time multi-user collaboration and automated itinerary generation.
- Integrated GPT-4o/Gemini with vector embeddings, reducing planning time by 75 percent.
- MERN-stack itinerary platform with JWT auth and multi-user collaboration.

Intelligent Salary Slip Automation Bot (RPA)

- Designed and deployed a UiPath-based RPA solution to automate end-to-end salary slip generation.
- Automated data extraction from Excel and dynamic template generation, producing personalized salary slips in under 3 minutes.
- Integrated a secure SMTP email system to distribute payslips to 30+ employees per cycle.

Technical Skills

Languages: C, Java, Python, JavaScript, SQL

Frameworks and Libraries: FastAPI, ReactJS, Next.js, Express.js, Tailwind CSS, PyTorch, RAG Pipelines, LLM Integration, Text Embeddings

Databases and Backend: Supabase, Qdrant, Vector Databases, MongoDB, Firebase, REST APIs, JWT Authentication

Tools and Platforms: Vercel, Streamlit, UiPath, Docker, Kubernetes, Jenkins, GitHub Actions, AWS

Certifications

Java Programming (NPTEL), DBMS (NPTEL), CCNA–Networks (Cisco), CCNA–Cybersecurity (Cisco)